Home Assignment - 2.

(1) Summarize challenges related to data velocity and variety and elaborate copying mechanisms.

(A) * Data Velocity challenge : Data is generated at very high speed, making real-time ingestion, processing, and storage difficult.

* Data Variety challenge :- Data arrives in multiple formats such as structured tables, JSON, text, images, audio and video.

Copying mechanisms :-

* Use streaming frameworks, distributed processing systems, flexible storage like data lakes and schema-on-read approaches to efficiently manage fast and diverse data.

(2) Describe challenges associated with unstructured data and elaborate strategies used to handle them.

Sol) Challenges :-

* Unstructured data lacks a predefined schema, making storage, indexing, searching and analysis complex and time-consuming.

Handling strategies :-

* Apply metadata tagging, indexing, natural language process and machine learning techniques to extract meaningful insights from unstructured data.

e) Illustrate qualitative and quantitative data and elaborate their usage in analytics applications.

i) Qualitative data :-

Non-numerical data such as text reviews, social media comments, and user feedback.

Quantitative data :-

Numerical data such as sales figures, temperature readings, and transaction counts.

Usage in analytics :-

* Qualitative data is used for sentiment and behavioral analysis, while quantitative data supports statistical analysis, trend identification, and predictive modelling.

f) Summarize challenges related to or involved in managing heterogeneous data sources and elaborate possible solutions.

A) Challenges :-

* Data comes from multiple sources with different formats, schemas, and update frequencies, causing integration and consistency issues.

Solutions :-

* Use standardized data formats, data integration tools, ETL/ELT pipelines, and centralized data platforms to unify and manage heterogeneous data.

) Classify data based on structure and usage and elaborate with examples.

(A) * Structured data : Organized in rows and columns commonly used in transactional systems and reporting.

* Semi-Structured data : JSON, XML, and logs, used in web applications and data exchange.

* Unstructured data : Text, images, audio, and videos, used in AI and machine learning applications.

(6) Describe the limitations in unstructured data processing and elaborate solutions.

(A) Limitations :-

* Difficult to query, analyze, and process due to lack of schema and high computational requirements.

Solutions :-

* Use distributed computing frameworks, indexing mechanisms, metadata management, and ML-based processing to improve efficiency.

(7) Illustrate numerical and non-numerical data and elaborate their analytical use.

(A) Numerical data : Enables mathematical operations, forecasting, performance analysis, and statistical modeling.

Non-Numerical data :-

Support pattern recognition, sentiment analysis, and user behavior understanding through advanced analytics.

(9) (13) Classify data types in real-time applications and elaborate their handling methods.

(A) Streaming data:- Continuously processed using real-time stream processing engines.

Event data :- Handled through message queues and event-driven architectures.

Sensor-data :- Managed using low-latency ingestion pipelines for immediate analysis and response.

(10) (14) Describe complexities of unstructured data storage and elaborate resolution techniques.

(A) Complexities :-

* Large data volume, lack of predefined structure, and difficulty in efficient data retrieval.

Resolution techniques:

* Use scalable object storage systems, data compression, metadata catalogs, and indexing for efficient storage and access.

(11) (15) Illustrate measurable and descriptive data and elaborate their analytical value.

(A) Measurable data:

Quantitative data that enables numerical comparison, performance evaluation, and predictive analytics.

Descriptive data:-

Qualitative data that provides context, explanations, and insights into trends and user behavior.

12. Summarize data integration challenges and elaborate solutions for real-time systems.

A.

Challenges :-

* High latency.
* Data inconsistency.
* Synchronization issues.
* Fault tolerance problems.

Solutions :-

* Implement real-time data integration pipelines, stream processing frameworks, scalable ingestion systems, and data validation mechanisms.